Chapter 1

Hybrid Sorghum Product Development and Production

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Abstract

There are many moving parts involved in developing and taking a new commercial hybrid to market. Sorghum [*Sorghum* bicolor (L.) Moench] hybrid development involves development of parental lines based on a cytoplasmic male sterile system including the pollen parent (R-line) and seed parents (A- and B-lines). New parental lines are developed by recombining existing elite parental lines to create new breeding populations or by adding specific traits of interest to existing parental lines by crossing elite lines with donor parents. Molecular markers are utilized to identify plants with particular traits of interest during parental line development. Newly developed parental lines are crossed together or with other elite parental lines to create new hybrid combinations. New hybrid combinations are evaluated in target geographies for improved yield, good agronomics, and specific traits of interest. Multi-year, multi-location evaluations are used to identify hybrid entries with improved yield, stable performance, and good agronomics. Evaluation of the parental lines involved in these new hybrids helps establish the produce ability and potential cost of goods which have direct impact on the potential commercial release of new hybrid products.

Key words Parental lines, A, B, R-lines, Elite, Hybrids, Molecular markers, Multi-year, Multi-location, Yield, Performance

1 Introduction

Developing and releasing a new sorghum hybrid into the market is a process that involves a variety of different stakeholders within a sorghum seed company. Sorghum breeders develop new parental lines which are evaluated as hybrid combinations in breeder trials. When potential new commercial products have been identified using data from breeder trials, breeder seed is provided to seedstock personnel for pilot hybrid seed production and parental line increase. Seed generated from these pilot production plantings is then provided to company agronomists and district sales managers for evaluation in field scale strip trials. Those strip trial entries with strong performance and good agronomics are in the next generation, produced by seed production staff for release into the market. Prior to release, company marketing personnel, using both breeding data and data from strip trials, position the new hybrid for

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release into a defined geography. Finally, sales personnel in the targeted release area introduce the new product to seed dealers, distributors, and customers.

The total process may take 10 to 12 generations from start to finish. By efficiently utilizing contra-season nurseries as well as greenhouse crops, which result in multiple generations per calendar year in the early line development stage, as well as conducting concurrent operations in the final hybrid evaluation and seed production stages, significant time can be shaved from this process. Access to genetic marker technology as well as technologies new to sorghum, e.g., double haploid, can also provide significant speed to market improvements within the process as described.

2 Materials

The first step in developing a new sorghum hybrid is parental line development. Sorghum has a perfect flower, as such, before entering into a description of hybrid sorghum product development; it is important to understand the mechanics of the cytoplasmic male sterile system utilized in the development and production of hybrid sorghums. Sorghum's cytoplasmic male sterile system is made up of three parental lines. The seed parent (female) has both sterile, A-line, and fertile, B-line, versions. The pollen parent (male), R-line, has restorer genes that restore fertility in F_1 hybrid combinations. A-lines, which are sterile, are propagated by crossing an A-line with a B-line of the same genotype. The B-line, sometimes referred to as the maintainer, is fertile and sheds pollen, but does not have restorer genes; thus progeny of A-line x B-line crosses is sterile. Both B-lines and R-lines are propagated by self-pollination. Commercial F_1 hybrids are generated by crossing A-lines x R-lines. The resulting progeny of an A x R cross is fertile due to the restorer genes provided by the R-line (pollen parent) in a hybrid combination.

3 Methods

3.1 Parental Line Development

The developmental process of seed parental lines (B-lines) and pollen parental lines (R-lines) is identical through the F_3 generation and begins with the process, whereby R-lines and B-lines are crossed/recombined with other R-lines or B-lines. As a general rule, R-lines are only crossed with R-lines and B-lines are only crossed with B-lines. In crops utilizing sterile systems, it is generally best to avoid mixing restorer lines and maintainer lines during parental line development. Elite R- and B-lines are chosen to be included in the recombination process based on their strong performance in hybrid combinations. On occasion, a particular parental line's inclusion into this recombination process might be the addition of a particular trait(s).

As both B-lines and R-lines that are to be crossed in the recombination process are fertile, it is necessary to emasculate one of the lines included in a combination in order to secure recombination from the other parent which provides the pollen. Emasculation is normally accomplished in one of two ways. One of the techniques called hand emasculation involves removing the anthers from 40 to 50 sorghum flowers prior to bloom by hand utilizing forceps or other small tools. This operation is quite tedious as the sorghum flowers are small and very delicate. Thus it is very easy to do damage to the flower if not done properly. The second technique involves using a plastic bag to cover a portion of the sorghum head prior to blooming. The plastic bag causes condensation under the bag and the moisture destroys the pollen when the anthers emerge during bloom. If done correctly, the hand emasculation method yields very few to no self-pollinations while the plastic bag method normally generates a higher frequency of self-pollinations. In both emasculation methods, pollen is provided by the other line involved in the combination and the resulting seed are a F_1 breeding population.

The F_1 seed generated following the emasculation process is then selfed. Care is taken to avoid harvesting any suspected self-pollinations which may have occurred during the emasculation process. Seed harvested from selfed F_1's is planted in F_2 nurseries at sites which are representative of target geographies. Available genetic markers are utilized to identify individual plants with specific traits of interest. From this subgroup which are positive for the markers of interest, selections are made by the breeder based on visual appearance and agronomic characteristics exhibited by individual plants. Population size in F_2 nurseries may vary depending on the trait(s) targeted, the number of genes involved, and the inherence of the trait(s). Minimum population size should be in the neighborhood of 1000 plants/segregating population, but a larger population size may be necessary depending on the elements involved in a particular population.

Selections made from segregating F_2 populations are planted in F_3 nurseries the following generation. Normally, a single row containing approximately 100 plants is planted from each selection. The presence of genetic markers for traits of interest is confirmed in each F_3 row. Breeder selections are then made at harvest from rows confirmed to be positive for markers of interest, and that have acceptable agronomic traits and good uniformity. Maturity is also an important consideration. Rows of elite parents are scattered in F_3 nursery plantings for the purpose of maturity comparisons.

3.2 R- and A-Line Development

While R-line and B-line populations are handled identically in generations F_1 through F_3, beginning with the F_4 generation selections planted in crossing nurseries are handled differently. The R-line F_4

selections are testcrossed onto two or three elite seed parents to generate F_1 hybrids which are utilized to evaluate the potential combinability of the new male parental lines (R-lines).

B-line F_4 selections are crossed on to elite-related females (A-lines) to begin the sterilization process ($CMS_{0\rightarrow1}$). By using a related female for the initial sterilization cross, testcrossing can normally occur with fewer sterile backcrosses. The following generation, CMS_1 A-line crosses are paired with the F_5 selections of the new B-line for an additional cross into sterile cytoplasm ($CMS_{1\rightarrow2}$). CMS_2 A-line crosses are paired with the F_6 selections of the new B-line in the next generation to provide a $CMS_{2\rightarrow3}$ version of the new A-line. Utilizing the same row of CMS_2 plants, testcrosses are also made with two or three elite pollen parents to generate F_1 hybrids which will be used to evaluate the potential combining ability of the new seed parents (A-lines). During the CMS process, sterility evaluations of potential new seed parents are monitored at each generation. Those entries with questionable sterility are discarded accordingly.

3.3 Testcross and Hybrid Evaluation

The testcross generation is the final step in parental line development. As discussed, there are quite a few generations required to get to the testcross stage. Contra season nurseries and greenhouse plantings are utilized to accomplish two to three generations per year during the early stages of parental line development as well as B-line sterilization during the CMS process.

After new parental lines are developed and testcrossed, the next step in the process is hybrid evaluation. New parental lines are initially evaluated as testcrosses. As mentioned previously, each new parental line, whether pollen parent (R-line) or seed parent (A-line), is testcrossed with elite tester lines. New pollen parents are crossed with two or three elite seed parents while new seed parents are crossed with two or three elite pollen parents. These testcrosses are evaluated in single rep trials at a minimum of three to four testing sites in target geographies. Adapted, maturity appropriate commercial check hybrids are included in these testcross trials. Based on yield performance relative to commercial check hybrids as well as maturity and agronomics, normally between 5 and 10% of the testcross entries are advanced to more intensive testing. New parental lines, both A-lines and R-lines, which have exhibited strong performance on multiple testers are flagged as potential general combiners (GCA). New parental lines, whether seed parents or pollen parents, which have shown GCA tendencies are crossed with a panel of additional elite lines to generate new F_1 hybrids. These hybrids by-pass the testcross evaluation stage and move directly into more intensive testing. Additionally, new parental lines flagged as possible GCA's move into a pool for recombination in order to generate new F_1 breeding populations for parental line development.

3.4 Advanced Hybrid Testing

Following the completion of testcross evaluation, in the next generation the 5–10% of testcross entries flagged for further consideration are advanced into more intensive testing and are evaluated in multi-rep, multi-location (5–6 sites) hybrid trials in target geographies. A panel of five to six well-adapted, maturity appropriate commercially available check hybrids, both internal and external, are included in each trial. Overall yield performance, maturity, stalk and root strength, as well as general agronomics are considered. The best hybrids relative to checks are moved into advanced level testing the next generation. Advance testing involves multi-rep, multi-location (10–12 sites) hybrid trials scattered over the target geographies. A panel of eight to ten maturity appropriate commercially available check hybrids both internal and external are included in each trial. A high priority is placed on total yield performance, which includes yield stability, yield for maturity, harvestable yield, as well as general agronomics.

Yield stability provides stable performance relative to competing products across environments and differing weather conditions which will vary from year to year. Generally those products that exhibit excellent yield stability tend to have wider adaptation. Yield for maturity identifies those hybrids that are able to generate the most yield with the earliest maturity in each maturity classification. A simple calculation of total yield divided by the number of days to mid-bloom is an excellent indicator of yield for maturity (yield/day). Harvestable yield takes into account all of the biotic and abiotic stressors which detract from the genetic yield potential that exists in a hybrid product. Insects, plant diseases, weak stalks or roots, drought stress, heat stress, and soil pH are all examples of stressors that can and do negatively impact harvestable yield.

3.5 Cost-of-Goods Evaluation

Concurrent with advanced testing, Cost-of-Goods (COG) evaluation of parental lines involved in those hybrids included in advance trials is initiated. Primary characteristics considered are seed parent (A-line) yield potential, pollen parent (R-line) pollen volume, and synchronization of bloom of the two parents involved in the hybrid. COG's evaluations are handled in replicated trials made up of parental lines utilized in advance trial entries. COG's trials are evaluated in seed production areas with planting dates commensurate with normal seed production plans. Seed parent yield potential has an obvious impact on cost of goods, but pollen volume and synchronization of bloom, commonly referred to as split, also impact green seed yields, as well as play a significant role in the genetic purity of the seed produced.

Pollen volume available during the period of female bloom is key to maintaining genetic purity and minimizing outcross potential. Low pollen volume R-lines are objectionable and once identified are discarded. Synchronization of bloom is controlled by split planting or sometimes referred to as delay plantings. The larger the difference in bloom timing between the seed parent and the pollen parent, the greater the risk of genetic purity issues. Split differences which require longer planting delays may be subject to weather issues which can alter planting plans. The goal of split planting is to have a large load of pollen available at the beginning of female bloom through the bloom period until the completion of bloom. It is common to make three male plantings to help insure good volumes of pollen availability during female bloom. The type of split plantings also has impact on risk of synchronization of bloom. Female delay plantings to achieve desired splits are generally accepted to be of higher risk than male delays.

3.6 Breeder's Seed Increase

During the same generation that advance testing occurs and COG's evaluation begins, breeder's seed increases of all new parental lines involved in hybrids evaluated as advance trial entries are initiated. Target quantities of these nursery breeders' seed increases normally are in the 5–10 lb. range. In the case of new seed parents, both A-line and B-line versions of the seed parent will need to be increased. In addition to seed of the A-line for pilot hybrid seed production, an increase in the A-line, as an AxB increase planted in one to two acres, will also be needed in order to be ready for a larger production of the new hybrid if warranted. The A-line and R-line increases are needed in order to be prepared for pilot seed production of those advance trial entries, which, based on performance, show promise as potential commercial releases. Seedstock personnel who grow these pilot hybrid seed production plantings (1 to 3 acres) normally have a target of 2500 to 5000 lbs. per hybrid. This seed will be utilized by company agronomists and district sales managers for field scale testing in strip trials the following growing season.

3.7 Strip Trials

Field scale strip trials with producers are scattered over the anticipated target release area. Strip trial entries include experimental hybrids as well as maturity appropriate commercially available hybrids for comparison. Data including yield, grain moisture, plant height, stalk quality, root strength, disease or insect response differences, as well as other differences that may manifest are collected and entered into a database that also includes small plot data collected from all breeding program evaluations involving this set of hybrids.

Experimental hybrids that exhibit yield potential improvement of 5% or greater vs. other maturity like products in the commercial hybrid product portfolio, stable performance relative to check hybrids across locations, good agronomic traits, and acceptable Cost-of-Goods are advanced to commercial status. These new offerings are normally targeted to replace a current product in the portfolio. The growing season prior to the planned sales of a new hybrid, seed production (500 to 1500 units) is planted to generate a targeted amount of the new hybrid product.

3.8 Seedstock Development

Seedstock of new parental lines which will be utilized by seed production staff is sourced from either new pollen parents harvested from pilot productions or in the case of new seed parents, seed produced in the one to two acre AxB increases mentioned previously. Planting split data from COG's evaluations as well as information generated from small pilot productions are used to determine planting splits for this introductory seed production. As mentioned, normally three planting dates of the pollen parent are utilized to help insure good pollen coverage during the seed parent bloom period.

3.9 Hybrid Seed Production

Hybrid sorghum seed production requires isolation from other grain sorghums, forage sorghums, sudangrass, and wild sorghum species including johnsongrass [*S. halepence* (L) Pers.] and shattercane [*S. bicolor* (L.) Moench]. Isolations can be generated with distance or with time utilizing delayed plantings. Following harvest of the seed production in the fall, growout samples drawn from harvested seed are included in winter growout plantings in Mexico, the Caribbean, or Central America to be evaluated for genetic purity prior to commercial sales. Following growout readings, those seed lots that are within company standards for genetic purity and germination are then conditioned and bagged.

4 Notes

Utilizing performance data generated from both breeder trials as well as field scale strip trial plantings, company marketing staffers position the new hybrid for release into a defined geography. As the final step in this process, sales personnel introduce the new product to seed dealers, distributors, and customers in the defined geography. The new product, which based on company-wide data is expected to have very strong performance in the targeted release area, is then anticipated to expand into other regions in subsequent growing seasons.

References

No references were utilized in preparing this chapter. Information in the chapter titled Hybrid Sorghum Development and Production was based on my reflections and observations during 40+ years of working in the private seed sector as a commercial plant breeder.